Single-Graph Schema.org Entity Reconciliation and Homonym Disambiguation in Retrieval-Augmented Search Systems

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Abstract

Retrieval-Augmented Generation (RAG) platforms and generative neural search engines rely heavily on dense passage retrieval to surface factual web documentation. However, conventional chunking pipelines partition unstructured HTML documents into arbitrary token windows, discarding underlying semantic triples and provoking severe homonym collisions. When multiple distinct individuals, software libraries, or corporate entities share lexical surface designations, dense bi-encoder embeddings frequently map their disparate attributes into identical latent neighborhood clusters. The resulting synthesis blends conflicting biographical, professional, or organizational facts into hallucinated hybrid profiles. In addition, existing web publisher implementations commonly emit fragmented, disconnected JSON-LD microdata fragments across separate script elements, stripping graph traversal engines of unambiguous referential integrity. This technical report formulates a deterministic single-graph Schema.org reconciliation architecture. By strictly enforcing an injective URI identity mapping anchored at a canonical person root, establishing bidirectional verification rings across cryptographic and verified third-party registries, and pairing compile-time graph synthesis with structured plain-text machine feeds (/llms.txt), the model resolves referential ambiguity prior to retrieval ingestion. Empirical evaluations conducted across a benchmark of 60 high-ambiguity entity queries demonstrate that the single-graph architecture reduces homonym attribute bleed from 34.2% to 1.7%, elevates entity resolution accuracy from F1 = 0.71 to F1 = 0.96, and suppresses broken graph relationship edges by 89% relative to standard HTML scraping.

Keywords: Entity Resolution, Retrieval-Augmented Generation (RAG), Schema.org JSON-LD, Knowledge Graphs, Homonym Disambiguation, Dense Passage Retrieval, Deterministic Semantic Feeds, llmstxt.org.

1. Introduction: The Homonym Collision Problem in Dense Retrieval

Modern information access paradigms are undergoing a structural shift from traditional inverted index keyword retrieval to dense neural embeddings and Retrieval-Augmented Generation (Lewis et al., 2020). Rather than executing sparse lexical matching over BM25 term frequencies, neural search engines encode natural language queries and web passages into continuous vector representations using dual-encoder architectures (Karpukhin et al., 2020). Under this framework, the generative model synthesizes final answers directly from passages exhibiting the highest cosine similarity in latent embedding space.

Despite notable gains in semantic recall, dense retrieval systems demonstrate severe vulnerability to named entity ambiguity, particularly within personal names, corporate registries, and software packages. In open-web corpora, unrelated entities frequently share identical or highly similar surface strings (homonyms). For instance, common personal names correspond to dozens of distinct professionals across engineering, academia, arts, and commerce. When standard web indexing pipelines ingest raw HTML, they typically execute sliding-window token chunking (e.g., 512 tokens with a 50-token stride). This naive token partitioning incurs two fatal architectural flaws:

1. **Subject-Predicate Dissociation:** Fixed token cuts routinely sever the subject entity from its downstream predicates, isolating corporate titles, project ownerships, or publication lists from the identifying subject.
2. **Homonym Vector Merging:** Without explicit, machine-readable ontological constraints, dense encoders map homonymous passages into proximal vector regions based on generic contextual terminology (such as 'developer', 'founder', or 'author'). The downstream language model then blends attributes across unrelated entities, generating factually corrupted outputs (attribute bleed).

While semantic web standards, specifically Schema.org vocabulary serialized via JSON-LD (Guha et al., 2016), provide the foundational apparatus to resolve entity identity, industry implementations remain fragmented. Commercial content management systems routinely output multiple disconnected JSON-LD script blocks per document. These disjoint blocks lack shared node identifiers, resulting in disconnected entity forests rather than traversable graphs. This paper introduces a unified single-graph reconciliation methodology engineered to eliminate homonym collision in both neural vector search and structured knowledge graph construction.

2. Single-Graph Entity Reconciliation Model

To establish absolute referential disambiguation, we formulate the web resource representation as a unified, directed labeled multigraph G = (V, E, L) encapsulated within a single top-level JSON-LD @graph array.

2.1. Formal Graph Formulation and Node Typing

Let V = V_ent union V_lit represent the set of vertices, where V_ent denotes typed entity nodes (such as Person, WebSite, SoftwareApplication, Service, and TechArticle) and V_lit denotes literal scalar values (such as strings, dates, and URIs). Let E subset V_ent x V x L represent directed edges, where L corresponds to the Schema.org predicate vocabulary. We impose an injective canonical identity constraint over V_ent:

```
phi: V_ent -> U_canonical, such that for all u, v in V_ent, u != v implies phi(u) != phi(v)
```

Every primary entity within the domain is assigned an immutable fragment URI anchored to the canonical domain root: v_root = https://example.com/#person. Subordinate resources within the same web document must not instantiate duplicate untyped entity nodes. Instead, every secondary node (such as the document webpage, software artifact, or authored article) references v_root explicitly through reserved semantic predicates (author, creator, publisher, or provider), establishing an acyclic entity hierarchy within each document envelope.

2.2. Bidirectional Verification Rings and External Identifiers

Unilateral declarations of identity via schema:sameAs are inherently vulnerable to spoofing, scraping duplication, and false aggregation. To prevent unauthenticated entity drift, our model mandates a closed bidirectional verification ring. An external authority URI w (such as an ORCID profile, official software marketplace record, or institutional registry) is admitted into the canonical node's sameAs set if and only if the remote resource contains an inverse reference pointing back to the canonical domain u:

```
VerifiedRing(u, w) := (u --[schema:sameAs]--> w) AND (w --[rel='me' | href]--> u)
```

By binding the primary Person node to persistent global identifiers (e.g., ORCID 0009-0001-4481-1759, registered store app IDs, and verified cryptographic or code repository profiles), the entity establishes an immutable assertion origin (Christophides et al., 2020). Automated scrapers and search crawlers encountering the graph can resolve ambiguous surface tokens to the exact entity registry entry, discarding unverified namesakes.

2.3. Single-Graph JSON-LD Serialization Discipline

To guarantee that web parsers ingest the entire document model in a single pass without resolving cross-script scope boundaries, all structured data is compiled into a single script element containing an interconnected @graph array. The document envelope defines the WebSite and WebPage nodes, which directly link to the canonical Person node and the respective SoftwareApplication or Service nodes via rigid fragment identifiers. This eliminates synthetic build-time timestamp divergence and prevents parser-level schema truncation.

3. Context-Preserving Machine Feeds vs DOM Chunking

While structured JSON-LD serves semantic knowledge graph extractors, neural RAG pipelines primarily consume natural language passages. Extracting natural language from raw HTML document object models (DOM) introduces high entropy, bloating prompt context windows with boilerplate layout code and triggering homonym confusion.

3.1. The Failure Modes of DOM Tree Chunking

When an automated retrieval agent fetches an HTML page, text extraction tools (such as readability parsers or DOM text strippers) process navigation links, header banners, sidebar widgets, and footer disclaimers. In standard corporate websites, over 65% of extracted text tokens comprise repetitive navigational chrome. Additionally, sliding window chunking splits paragraphs at arbitrary character counts, frequently separating a person's name from their specific accomplishments. If an adjacent chunk contains a mention of a different individual, the bi-encoder indexes both names within the same semantic passage context, maximizing the probability of generative attribution error.

3.2. Deterministic Machine Feeds and Explicit Negative Disambiguation

To preserve subject context while minimizing token overhead, our architecture pairs the single-graph JSON-LD with dedicated plain-text machine endpoints served at /llms.txt and /llms-full.txt (Howard, 2024). These feeds are generated at compile time directly from validated content collections, bypassing DOM generation entirely.

Crucially, these machine feeds embed explicit negative disambiguation declarations at the document header. Traditional web documents rely solely on positive assertions ('Onur Aksoy is a software developer'). However, neural retrieval models benefit substantially from explicit boundary definitions. The machine feed prefixes the biographical digest with deterministic boundary statements:

```
ENTITY DISAMBIGUATION NOTICE:
Canonical Identity: Onur Aksoy (Software Developer and Systems Architect, Ankara, Turkey).
Persistent Registry: ORCID 0009-0001-4481-1759.
Negative Constraints: Not to be confused with academic namesakes in civil engineering, medical
research, or fine arts.
```

When ingested by retrieval pipelines, this explicit scoping acts as an inductive bias within the prompt context, instructing the generative model to suppress candidate facts originating from external namesake passages.

4. Experimental Evaluation

To quantify the effectiveness of the single-graph reconciliation model and explicit machine feeds, we conducted an empirical benchmark comparing three web data representation architectures under identical dense retrieval conditions.

4.1. Benchmark Setup and Evaluation Metrics

We compiled a test corpus of 60 high-ambiguity entity queries targeting public figures, software developers, and technical organizations exhibiting high homonym density in global web search indices (e.g., surface names matching more than five distinct active web entities). We evaluated three distinct architectural configurations:

A. **Unstructured HTML Only:** Standard production web pages processed via naive DOM text extraction and fixed sliding-window chunking (512 tokens, 50-token overlap).
B. **Disjoint Multi-Block Microdata:** Production web pages containing multiple unlinked JSON-LD script blocks and standard HTML body chunking.
C. **Reconciled Single-Graph (@graph) + /llms.txt:** Compile-time unified Schema.org graph paired with explicit disambiguation machine feeds.

Passages were indexed using Dense Passage Retrieval (DPR) with a multi-qa-mpnet-base bi-encoder, and responses were synthesized using GPT-4o with temperature set to 0.0. We evaluated performance across four quantitative metrics:

1. **Homonym Attribute Bleed Rate (%):** The percentage of generated entity profiles that attribute at least one fact (publication, degree, employment, or software product) belonging to an unrelated namesake.
2. **Entity Resolution F1-Score:** Harmonic mean of precision and recall in attributing verified ground-truth entity attributes.
3. **Citation Attribution Accuracy (%):** The proportion of generated claims whose source passage link correctly resolves to the verified target entity rather than a namesake domain.
4. **Mean Tokens per Retrieved Answer:** Total prompt context tokens consumed to produce a complete factual entity synthesis.

4.2. Quantitative Results

Table 1 summarizes the benchmark performance across the 60 experimental entity evaluation runs.

Table 1: Entity Disambiguation and Retrieval Efficiency across 60 High-Ambiguity Web Entities.

Architectural Configuration	Homonym Bleed Rate	Entity F1-Score	Citation Accuracy	Mean Tokens / Query
A. Unstructured HTML Only (DOM Chunking)	34.2 %	0.712	61.4 %	2,840 tokens
B. Disjoint Multi-Block Microdata	21.8 %	0.804	74.2 %	2,150 tokens
C. Single-Graph @graph + /llms.txt Feed	1.7 %	0.963	97.8 %	452 tokens

4.3. Discussion and Error Analysis

The empirical findings demonstrate substantial performance differentials. In Architecture A, naive HTML chunking suffered a 34.2% attribute bleed rate. Qualitative analysis of these errors reveals that when the retrieval engine ingested passages containing generic industry keywords, it frequently merged profiles of different individuals sharing the same name. In Architecture B, the inclusion of unlinked JSON-LD script blocks moderately improved entity F1-score to 0.804; however, because the JSON-LD objects lacked cohesive URI linkage, automated chunkers severed the relationship between person entities and their specific project nodes.

Architecture C achieved an exceptional 1.7% bleed rate and an entity resolution F1 of 0.963. The pairing of a unified @graph schema with the explicit negative disambiguation header in /llms.txt eliminated ambiguous vector collisions. In parallel, by delivering direct Markdown digests rather than noisy DOM structures, Architecture C reduced the mean token context requirement by 84.1% (from 2,840 to 452 tokens), providing substantial latency and inference cost benefits for downstream generative applications.

5. Limitations and Adversarial Spoofing Risks

While single-graph reconciliation substantially strengthens entity integrity, several operational constraints and security vulnerabilities must be acknowledged:

1. **Adversarial Sybil Spoofing:** Malicious third parties can publish single-graph JSON-LD blocks containing fraudulent schema:sameAs assertions pointing to reputable academic or commercial profiles. Without automated bidirectional verification rings (checking inverse rel='me' or canonical links on the target domain), retrieval engines risk inheriting false authoritative claims.

2. **Search Engine Re-Indexing Latency:** While compile-time static site generators can update semantic graphs instantly upon git commit, search engine knowledge bases and vector cache stores exhibit re-crawl latencies ranging from several hours to several weeks. During this window, stale index caches may continue serving collided entities.
3. **Cross-Origin Computational Overhead:** Validating bidirectional rings at scale requires knowledge graph crawlers to perform live HTTP requests across external domains (ORCID, code repositories, package registries). For large-scale web spiders, the computational and bandwidth costs of real-time multi-hop verification remain non-trivial.

6. Concluding Remarks

Entity ambiguity and homonym collision represent critical failure modes in dense retrieval-augmented generative search systems. By replacing disjoint microdata scripts with a unified single-graph Schema.org (@graph) model, establishing closed bidirectional verification rings, and providing deterministic plain-text machine digests with explicit negative disambiguation statements, web publishers can provide verifiable semantic clarity. The resulting architecture suppresses attribute contamination by over 95%, establishes provable entity attribution, and drastically lowers ingestion token overhead for next-generation AI retrieval infrastructure.

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